

Circuit Theory

Chapter 11

AC Power Analysis

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

AC Power Analysis

Chapter 11

- 11.1 Instantaneous and Average Power
- 11.2 Maximum Average Power Transfer
- 11.3 Effective or RMS Value
- 11.4 Apparent Power and Power Factor
- 11.5 Complex Power
- 11.6 Conservation of AC Power
- 11.7 Power Factor Correction
- 11.8 Power Measurement

11.1 Instantaneous and Average Power (1)

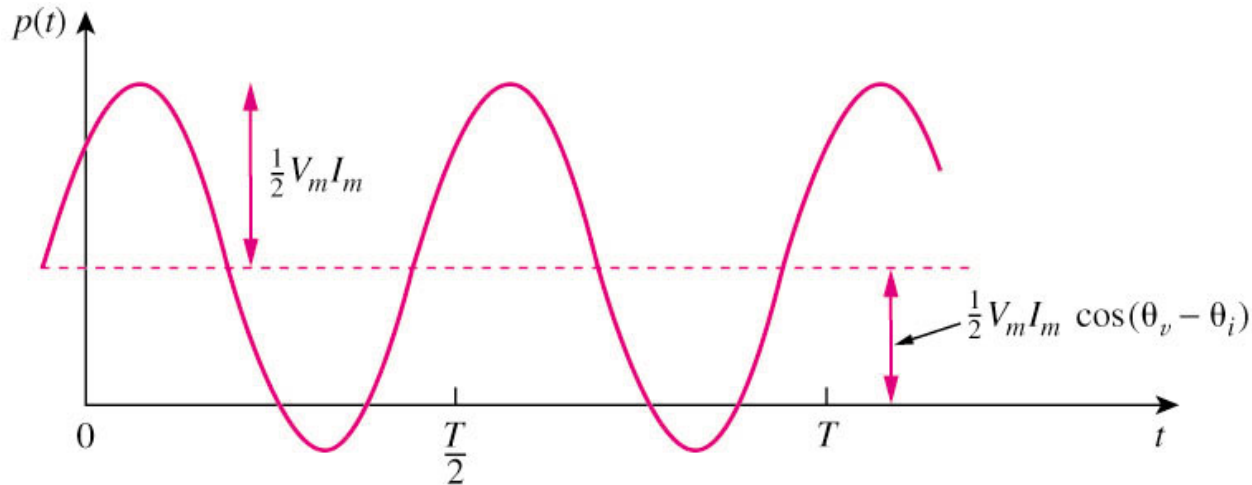
- The instantaneous power, $p(t)$

$$p(t) = v(t) i(t) = V_m I_m \cos(\omega t + \theta_v) \cos(\omega t + \theta_i)$$

$$= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) + \frac{1}{2} V_m I_m \cos(2\omega t + \theta_v + \theta_i)$$

Constant power

Sinusoidal power at $2\omega t$

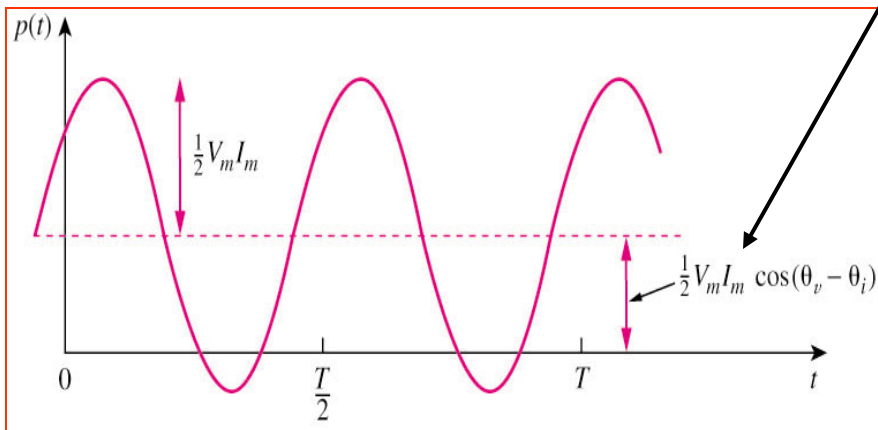


$p(t) > 0$: power is absorbed by the circuit; $p(t) < 0$: power is absorbed by the source.

11.1 Instantaneous and Average Power (2)

- The average power, P , is the average of the instantaneous power over one period.

$$P = \frac{1}{T} \int_0^T p(t) dt = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i)$$



- P is not time dependent.
- When $\theta_v = \theta_i$, it is a purely resistive load case.
- When $\theta_v - \theta_i = \pm 90^\circ$, it is a purely reactive load case.
- $P = 0$ means that the circuit absorbs no average power.

11.1 Instantaneous and Average Power (3)

Example 1

Calculate the instantaneous power and average power absorbed by a passive linear network if:

$$\begin{aligned}v(t) &= 80 \cos(10t + 20^\circ) \\ i(t) &= 15 \cos(10t + 60^\circ)\end{aligned}$$

Answer: $385.7 + 600\cos(20t - 10^\circ)\text{W}$, 387.5W

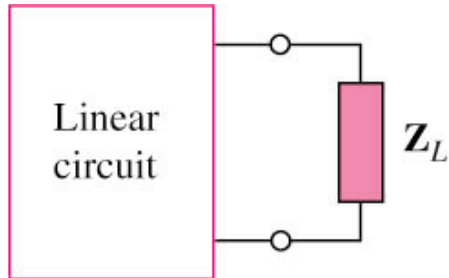
11.1 Instantaneous and Average Power (4)

Example 2

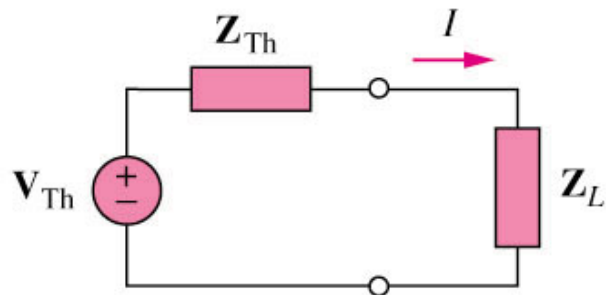
A current $\mathbf{I} = 10 \angle 30^\circ$ flows through an impedance $\mathbf{Z} = 20 \angle -22^\circ \Omega$. Find the average power delivered to the impedance.

Answer: 927.2W

11.2 Maximum Average Power Transfer (1)



(a)



(b)

$$Z_{TH} = R_{TH} + jX_{TH}$$

$$Z_L = R_L + jX_L$$

The maximum average power can be transferred to the load if

$$X_L = -X_{TH} \text{ and } R_L = R_{TH}$$

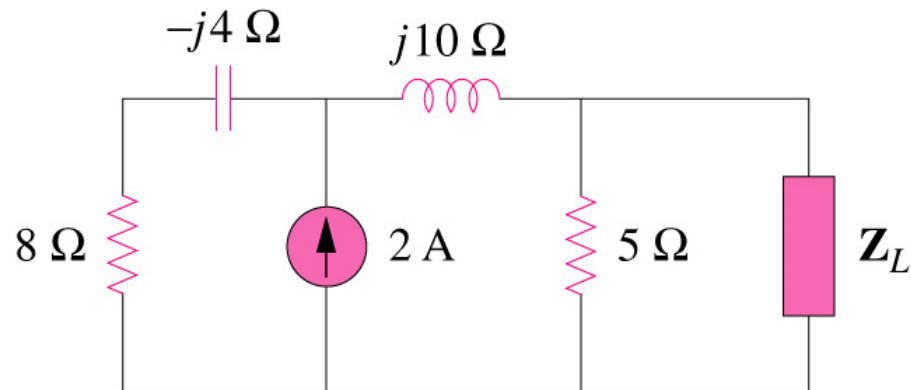
$$P_{\max} = \frac{|V_{TH}|^2}{8 R_{TH}}$$

If the load is purely real, then $R_L = \sqrt{R_{TH}^2 + X_{TH}^2} = |Z_{TH}|$

11.2 Maximum Average Power Transfer (2)

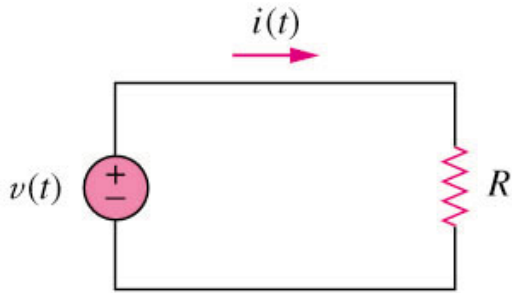
Example 3

For the circuit shown below, find the load impedance Z_L that absorbs the maximum average power. Calculate that maximum average power.



Answer: $3.415 - j0.7317\ \Omega$, 1.429 W

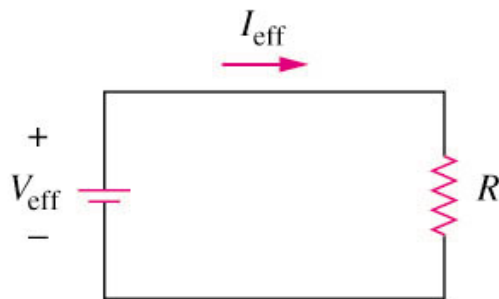
11.3 Effective or RMS Value (1)



(a)

The total power dissipated by R is given by:

$$P = \frac{1}{T} \int_0^T i^2 R dt = \frac{R}{T} \int_0^T i^2 dt = I_{rms}^2 R$$



(b)

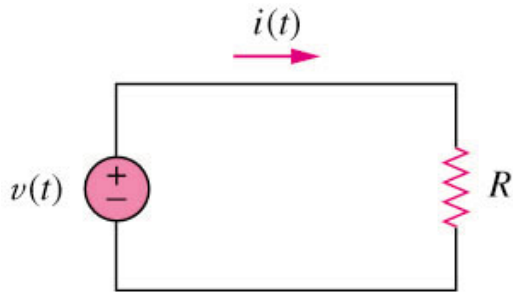
Hence, I_{eff} is equal to:

$$I_{eff} = \sqrt{\frac{1}{T} \int_0^T i^2 dt} = I_{rms}$$

The rms value is a constant itself which depending on the shape of the function $i(t)$.

The effective of a periodic current is the dc current that delivers the same average power to a resistor as the periodic current.

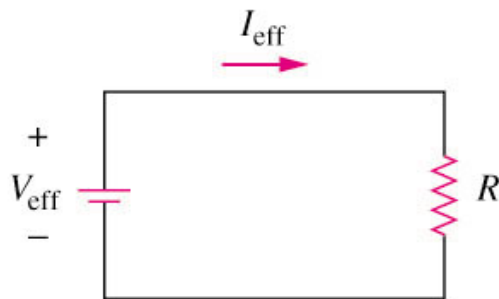
11.3 Effective or RMS Value (2)



(a)

The rms value of a sinusoid $i(t) = I_m \cos(\omega t)$ is given by:

$$I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$



(b)

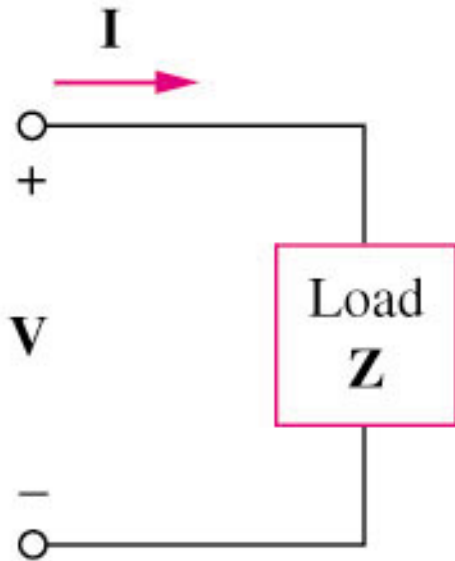
The average power can be written in terms of the rms values:

$$P_{\text{eff}} = \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i)$$

Note: If you express amplitude of a phasor source(s) in rms, then all the answer as a result of this phasor source(s) must also be in rms value.

11.5 Complex Power (1)

Complex power **S** is the product of the voltage and the complex conjugate of the current:

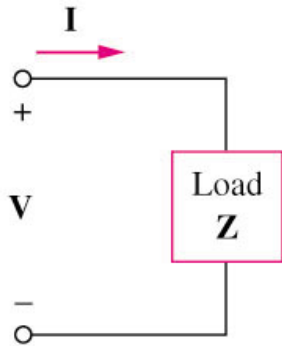


$$V = V_m \angle \theta_v$$

$$I = I_m \angle \theta_i$$

$$\frac{1}{2} V I^* = V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i$$

11.5 Complex Power (2)



$$S = \frac{1}{2} V I^* = V_{\text{rms}} I_{\text{rms}} \angle \theta_v - \theta_i$$

$$\Rightarrow S = V_{\text{rms}} I_{\text{rms}} \cos(\theta_v - \theta_i) + j V_{\text{rms}} I_{\text{rms}} \sin(\theta_v - \theta_i)$$

$$S = P + jQ$$

P: is the average power in watts delivered to a load and it is the only useful power.

Q: is the reactive power exchange between the source and the reactive part of the load. It is measured in VAR.

- $Q = 0$ for *resistive loads* (unity pf).
- $Q < 0$ for *capacitive loads* (leading pf).
- $Q > 0$ for *inductive loads* (lagging pf).